# END STATE HIGH LEVEL ARCHITECTURE (PROCESS VIEW)

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# EXECUTIVE SUMMARY

- This deck outlines RS Group's **logical end-state enterprise architecture** across the five major value streams: O2C, P2F, P2A, R2R and S2P. *(physical deployment and NFRs will be defined in later phases)*

- Each tower is shown through a Process View and Application View, providing a clear view of how future business processes and systems will operate.

- RS Group will adopt an **S/4HANA Greenfield Implementation**, as only ~25–30% of ECC objects map cleanly to S/4Hana (**takes into consideration Plants Abroad requirement**).

- SAP MDG* will act as the **enterprise master data hub** to ensure clean, governed and consistent data for all markets.

- The End-State Logical Architecture positions S/4HANA as the digital core, supported by:

  - Master Data & Governance: SAP MDG / Informatica*
  - Planning & Consolidation: OneStream, SAP SAC (BDC), SAP Datasphere (BDC), Snowflake and PowerBI for analytics, Blue Yonder
  - Data & Analytics: SAP Datasphere (BDC), Snowflake, PowerBI
  - Execution & Compliance: SAP Ariba*, iCertis*, SAP Vistex, SAP aATP, SAP EWM, SAP DRC, SAP Vertex, SAP Blackline*, SAP Multibank Connectivity, OpenText

**The architecture is built on key principles:**

- Clean and governed master data
- Simplified, rationalised application landscape
- Retirement of ECC, BW, DI and tactical tools
- Standardised global processes and templates
- Cloud-native analytics

***This delivers RS Group's North Star architecture: a modern, scalable and future-ready platform enabling consistent operations, better insight and strong foundations for automation and AI.***



* is still to be confirmed & pattern to be agreed

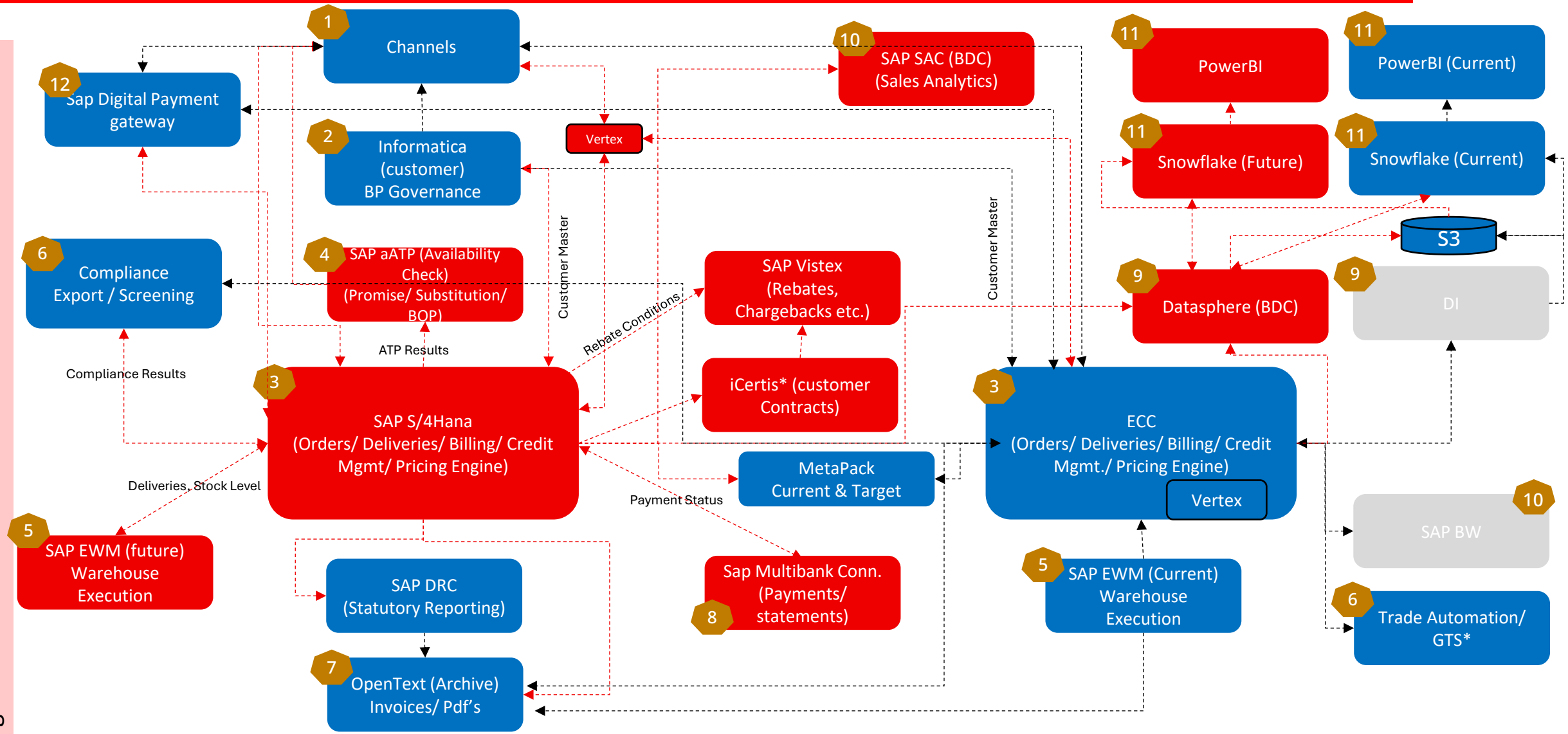
Internal

# Order to Cash :

# Application View

# O2C END-STATE ARCHITECTURE OVERVIEW (HIGH LEVEL)

Flow: Order Capture → Availability → Execution → Billing → Reporting




- This diagram shows > How customer orders move through our future systems.
- Salesforce captures customers and quotes
- S/4HANA becomes the central engine for orders, deliveries and billing. aATP checks product availability,
- EWM handles warehouse execution, and Vistex manages rebates.
- GTS*/ Trade Automation ensures compliance, OpenText stores invoices, and Multi-Bank Connectivity manages payments. Snowflake and PowerBI provide sales reporting, while ECC and legacy tools are gradually phased out.

* is still to be confirmed & pattern to be agreed

Planned for deprecation | Tactical Solution | End State | Current

Current →

Future →

# O2C Flow summarised

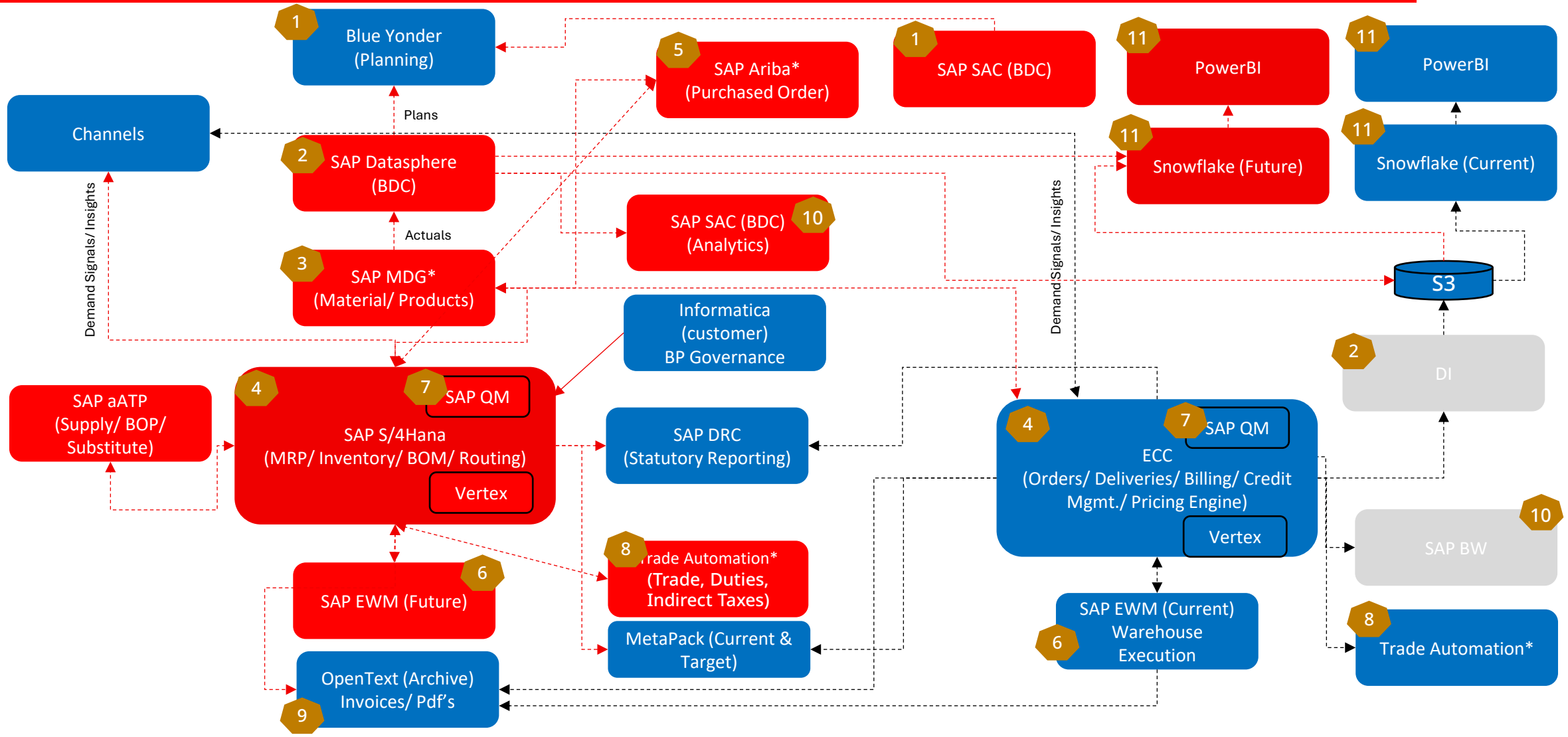
| No. | System | Details |
|---|---|---|
| 1 | Channels | Quotes, opportunities, customer creation, Front End |
| 2 | Informatica | Customer master governed |
| 3 | S/4HANA Sales or ECC | Orders → Deliveries → Billing |
| 4 | aATP | Availability check, supply match, BOP |
| 5 | SAP EWM (Current or Future) | Picking, packing, dispatch |
| 6 | Trade Automation/GTS* | Export compliance, ESG & Supplier screening (TBC). The usage of SAP GTS or Trade Automation needs to be confirmed |
| 7 | OpenText | Invoice & delivery document archive |
| 8 | SAP Multibank Connectivity | Payment receipts, statements |
| 9 | SAP DataSphere or SAP DI | Sales & fulfilment actuals |
| 10 | Sap BW or SAP SAC | Analytics |
| 11 | Snowflake → PowerBI | Commercial analytics, O2C KPIs (this will cover both current ECC & future S/4 |
| 12 | SAP Payment gateway | To process digital payment |

# Plan to Fulfil :

# Application View

# P2F END-STATE ARCHITECTURE OVERVIEW (HIGH LEVEL)

Flow: Demand Planning → Supply Planning → MRP → Production / Procurement → Warehouse Execution → Shipment → Reporting




This slide shows how we plan, schedule, and fulfil products in the future.

- Blue Yonder creates the supply and demand plans
- Datasphere brings all the data together and S/4HANA runs MRP, inventory, and production.
- aATP checks availability
- EWM manages warehouse execution and QM handles quality.
- GTS ensures compliance and OpenText manages documents.
- Snowflake and PowerBI give us full supply chain visibility, while older systems like ECC and BW are phased out.

* is still to be confirmed & pattern to be agreed

Planned for deprecation | Tactical Solution | End State | Current

Current →

Future →

# P2F Flow summarised

| No. | System | Details |
|---|---|---|
| 1 | Blue Yonder + SAP SAC | Demand / supply plans generated |
| 2 | SAP Datasphere or SAP DI | Planning models harmonised & fed downstream/ upstream |
| 3 | SAP MDG | Product & material governance |
| 4 | S/4HANA Supply Chain or ECC | MRP → Production Orders → Procurement triggers |
| 5 | SAP Ariba | Purchase Orders |
| 6 | SAP EWM (Current or Future) | Warehouse execution for stock movements |
| 7 | SAP QM | Inspection lots, quality results |
| 8 | Trade Automation* | Trade compliance (if export scenario). The usage of SAP GTS or Trade Automation needs to be confirmed |
| 9 | OpenText | Packaging labels, shipment docs |
| 10 | Sap BW or SAP SAC | Analytics |
| 11 | Snowflake → PowerBI | Commercial analytics, O2C KPIs |

# BIG-ROCK VS INCREMENTAL ADOPTION

The Big-Rocks define our future Plan-to-Fulfil operating model.
Everything else can be phased once the foundation is in place.

**Big-Rock Capabilities (Foundational – Decide Once)**

| Capability | What it Anchors in P2F |
|---|---|
| S/4HANA as Supply Chain Core | Single execution backbone for MRP, inventory, production & fulfilment |
| Planning → Execution Integration | Blue Yonder plans driving S/4HANA MRP & procurement |
| Advanced ATP (aATP) | Customer promise dates, allocations & backorder logic |
| Target Warehouse Model (SAP EWM) | One warehouse execution & stock-truth model |
| Master Data Governance (SAP MDG)* | Single product/material definition across planning & execution |
| Trade, Tax & Statutory Architecture | Compliance consistency (GTS / Vertex / DRC) |
| Analytics Backbone (SAC + Snowflake) | One KPI definition for OTIF, inventory, service & margin |

**Incremental Capabilities (Phased – Optimise Over Time)**

| Capability | How it Evolves |
|---|---|
| Network & Safety Stock Optimisation | Gradual maturity by region / DC |
| Consumption-Based Replenishment | SKU-by-SKU adoption |
| Subcontracting Scenarios | Vendor-specific rollout |
| Cross-Docking | DC-by-DC enablement |
| Advanced Quality Management | Product criticality driven |
| Supplier Collaboration (Ariba*) | Supplier – by – supplier |
| Advanced PowerBI Dashboards | KPI rollout time |

# Plan-to-Fulfil End-State: Process Walkthrough (Business View)

# SAP S/4HANA – PLAN-TO-FULFIL (P2F) STANDARD SCENARIOS - 1

| P2F Phase | # | Scenario/ Use Case | SAP Description | Mapped in architecture |
|---|---|---|---|---|
| Plan | 1 | Demand Forecasting | Statistical & consensus demand planning using historical data and overrides | Blue Yonder + SAP SAC (BDC) |
| Plan | 2 | Supply planning | Capacity-aware supply plans across plants/DCs | Blue Yonder → S/4HANA |
| Plan | 3 | Network Planning | Multi-node network balancing & safety stock optimisation | Blue Yonder |
| Plan | 4 | Rough-Cut Capacity Planning | High-level capacity validation before execution | Blue Yonder + SAC |
| Core S/4 | 5 | Material Master Governance | Controlled creation and replication of material/product data | SAP MDG* |
| Core S/4 | 6 | BOM & Routing Governance | Governance of BOMs, routings and production versions | S/4Hana |
| PLAN → EXECUTE | 7 | MRP Live | Net requirements planning generating PRs & planned orders | S/4Hana |
| PLAN → EXECUTE | 8 | Consumption-Based Replenishment | Reorder point, forecast-based or min/max planning | S/4Hana |
| PLAN → EXECUTE | 9 | Advanced ATP (aATP) | Product allocation, substitution & backorder processing | SAP aATP |
| Source | 10 | External Procurement (Stock) | PR → PO → GR → Inventory update | SAP Ariba* + S/4HANA |
| Source | 11 | Subcontracting | Components issued to vendor and finished goods received | S/4Hana |
| Source | 12 | Third-Party / Drop Shipment | Vendor ships directly to customer | Channels → S/4HANA |

# SAP S/4HANA – PLAN-TO-FULFIL (P2F) STANDARD SCENARIOS - 2

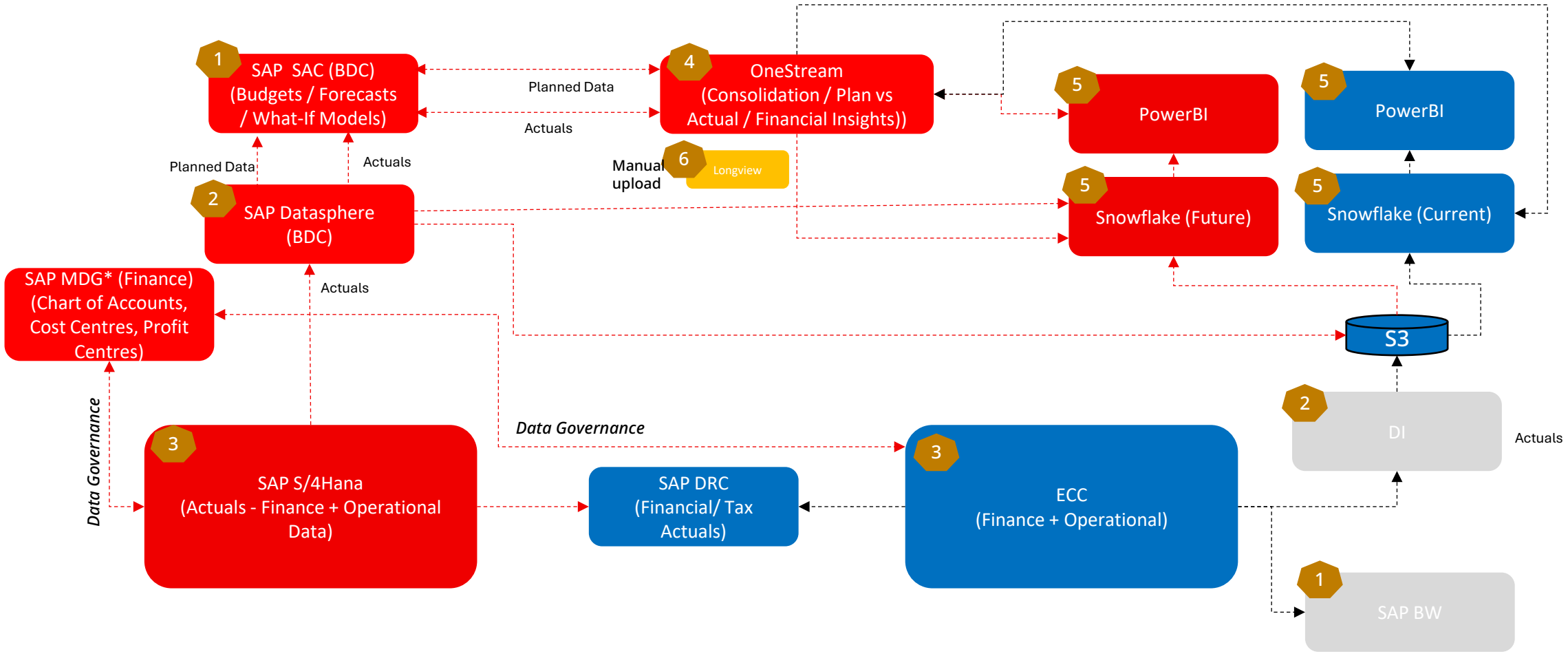
| P2F Phase | # | Scenario/ Use Case | SAP Description | Mapped in architecture |
|---|---|---|---|---|
| Make | 13 | Discrete Manufacturing | Order-based production execution | S/4Hana |
| Make | 14 | Repetitive Manufacturing | Rate-based, high-volume manufacturing | S/4Hana |
| Make | 15 | Quality-Managed Production | In-process & GR inspections with usage decisions | SAP QM |
| Move | 16 | Warehouse Execution | Inbound, putaway, picking & staging | SAP EWM |
| Move | 17 | Cross-Docking | Direct flow from inbound to outbound | SAP EWM |
| Move | 18 | Internal Stock Transfers | Plant ↔ plant / DC ↔ DC movements | S/4Hana + SAP EWM |
| Deliver | 19 | Make-to-Stock Fulfilment | SO → ATP → Delivery → PGI → Billing | S/4Hana |
| Deliver | 20 | Make-to-Order Fulfilment | Sales order triggers production | S/4Hana |
| Deliver | 21 | Intercompany Fulfilment | Internal sales with transfer pricing | S/4Hana |
| Comply | 22 | Trade Compliance | Sanctions, customs, duty determination | SAP GTS*/ Trade Automation* |
| Comply | 23 | Indirect Tax Calculation | VAT/GST determination | Vertex |
| Document | 24 | Logistics Documentation | Labels, packing lists, shipment docs | OpenText |
| Monitor | 25 | Supply Chain Operational Analytics | Inventory turns, service level, backorders | SAP SAC (BDC) |
| Monitor | 26 | Commercial & O2C Analytics | Margin, OTIF %, commercial KPIs | Snowflake → PowerBI |
| Regulate | 27 | Statutory & Digital Reporting | SAF-T, e-Invoicing, country reporting | SAP DRC |

# Plan to Actual :

# Application View

# P2A END-STATE ARCHITECTURE OVERVIEW (HIGH LEVEL)

Flow: Plan → Execute → Record Actuals → Consolidate → Analyse → Report




This slide shows how we bring together planned data and actual data to measure business performance.

- SAC creates our plans
- S/4 provides actuals and Datasphere harmonises the data.
- OneStream supports consolidation and plan-vs-actual analysis, while Snowflake and PowerBI provide enterprise-level reporting and insights.
- Legacy systems like ECC and BW are phased out as we move to this modern planning and analytics architecture.

* is still to be confirmed & pattern to be agreed

Planned for deprecation | Tactical Solution | End State | Current

Current →

Future →

# P2A Flow summarised

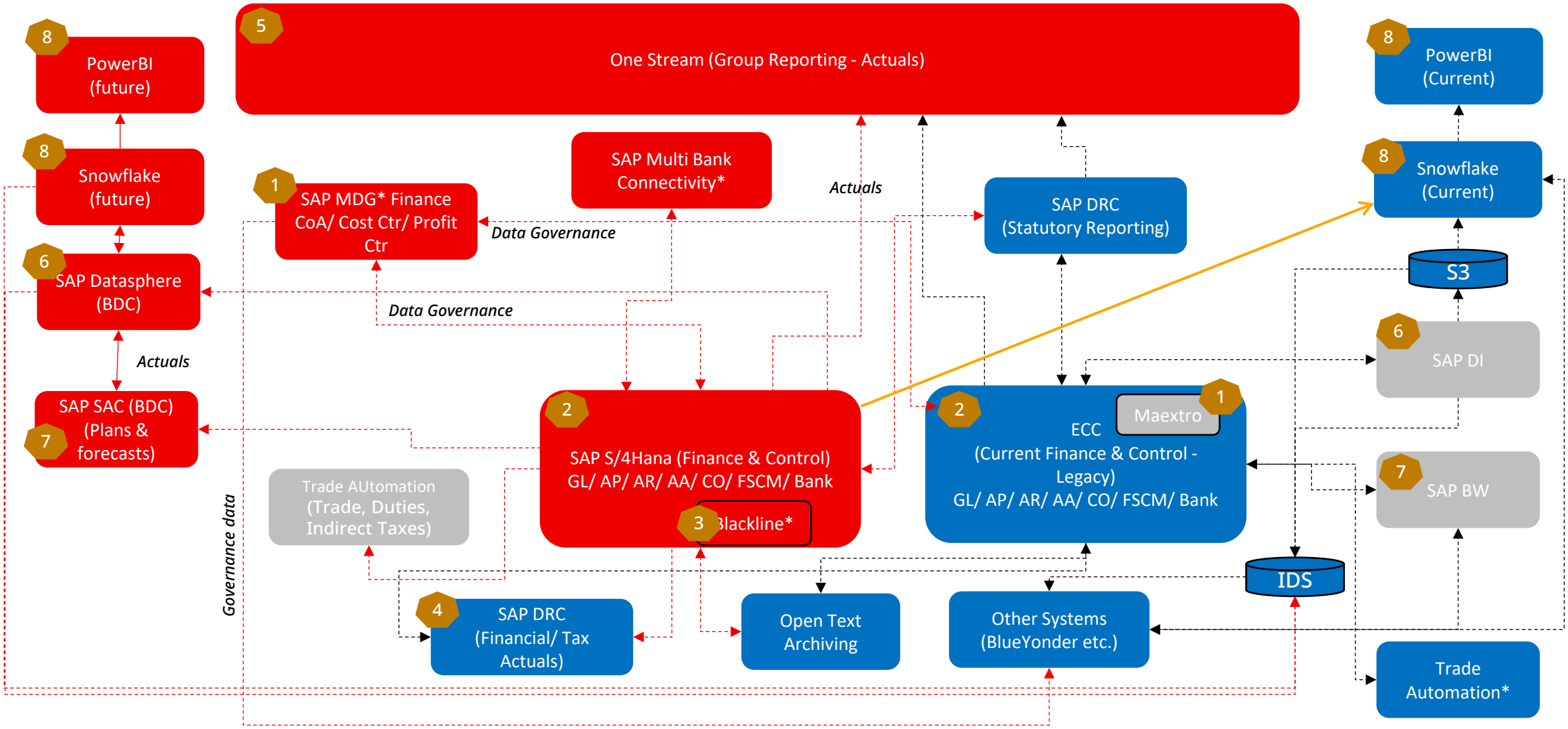
| No. | System | Details |
|---|---|---|
| 1 | SAP SAC or SAP BW | Budgets, forecasts, scenarios |
| 2 | SAP Datasphere or SAP SAP DI | Plan data harmonised |
| 3 | S/4HANA (Actuals) or ECC (Actuals) | Financial & operational actuals posted |
| 4 | OneStream | Consolidation + plan vs. actual variance |
| 5 | Snowflake → PowerBI | Commercial analytics, O2C KPIs |
| 6 | Longview | Longview (shown as tactical, will be reviewed once S/4 is completed) used for Group Tax reporting (currently no integration planned. |

# Record to Report :

# Application View

# R2R END-STATE ARCHITECTURE OVERVIEW (HIGH LEVEL)

Flow: MDG Finance → S/4 Finance → BlackLine → DRC → OneStream → Datasphere → Snowflake → PowerBI




- S/4HANA becomes the single place where all financial data lives.
- SAP MDG* keeps our data clean, OneStream handles group reporting, and SAC/Datasphere help us plan and analyse.
- Snowflake and PowerBI provide wider reporting.
- Older systems like ECC and BW will be phased out as we move to this new setup."

* is still to be confirmed & pattern to be agreed

Planned for deprecation | Tactical Solution | End State | Current

Current → | Future →

# R2R Flow summarised

| No. | System | Details |
|---|---|---|
| 1 | SAP MDG* or Sap Maextro | Product & material governance |
| 2 | S/4HANA Supply Chain or ECC | MRP → Production Orders → Procurement triggers |
| 3 | SAP Blackline | Account reconciliation & financial close |
| 4 | SAP DRC | Statutory & compliance reporting |
| 5 | One Stream | Group reporting & financial consolidation |
| 6 | SAP Datasphere or SAP DI | Harmonised Financial Model |
| 7 | Sap BW or SAP SAC | Analytics |
| 8 | Snowflake → PowerBI | Commercial analytics, O2C KPIs |

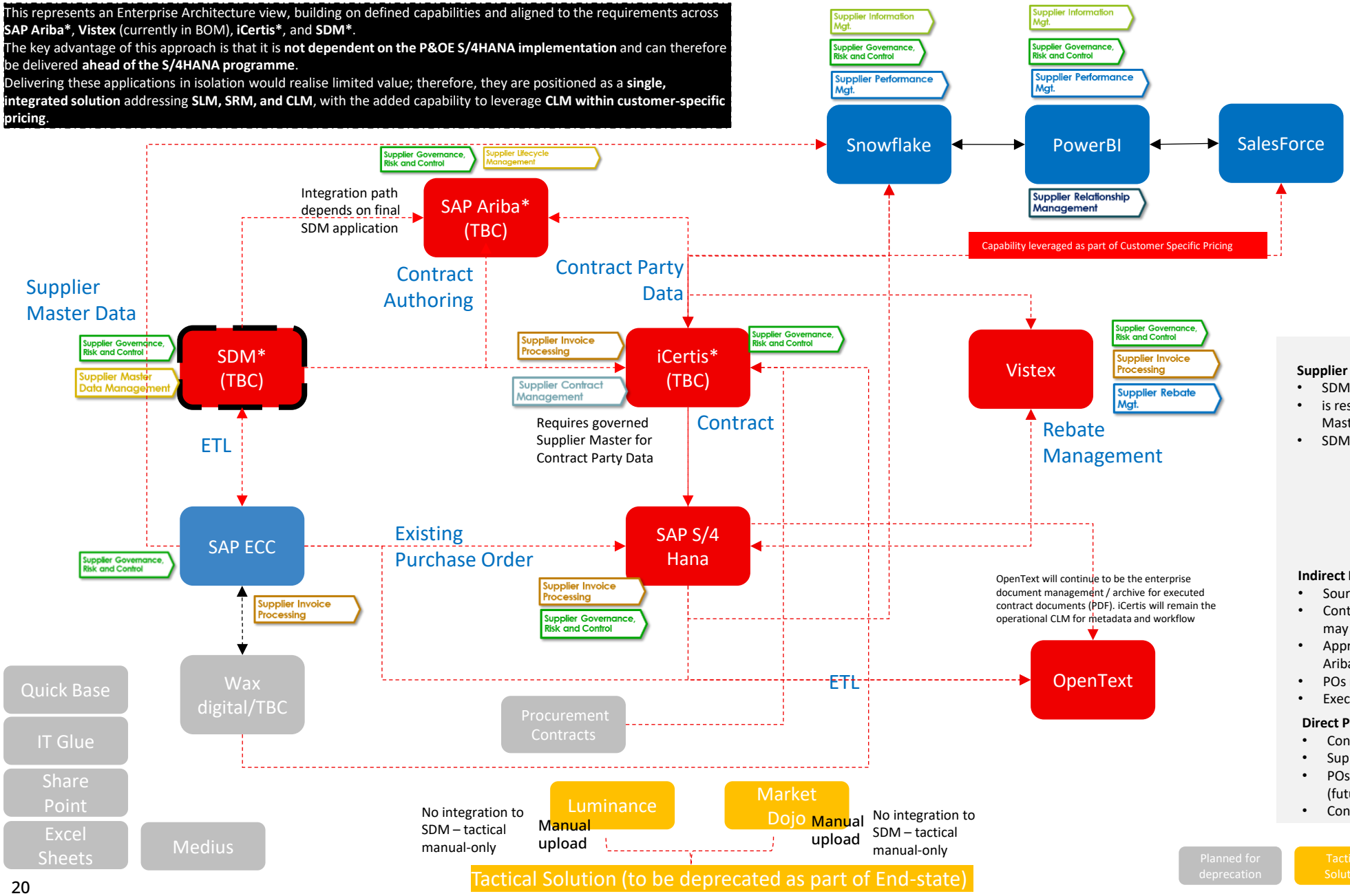
Internal

# Source to Pay :

# CLM/ SRM – Application View

# END-STATE S2P ARCHITECTURE OVERVIEW (HIGH LEVEL)

This represents an Enterprise Architecture view, building on defined capabilities and aligned to the requirements across **SAP Ariba***, **Vistex** (currently in BOM), **iCertis***, and **SDM***.
The key advantage of this approach is that it is **not dependent on the P&OE S/4HANA implementation** and can therefore be delivered **ahead of the S/4HANA programme**.
Delivering these applications in isolation would realise limited value; therefore, they are positioned as a **single, integrated solution** addressing **SLM, SRM, and CLM**, with the added capability to leverage **CLM within customer-specific pricing**.




***Note:***
- ✓ *Supplier Data Management (SDM) is shown as a capability. The application to deliver SDM is TBC. The final choice must support:*
  - ✓ *SAP S/4HANA Business Partner data model*
  - ✓ *Multilingual supplier semantics*
  - ✓ *Supplier lifecycle governance*
  - ✓ *Integration with Ariba, iCertis, Vistex, ECC and S/4*
- ✓ *SAP Ariba and iCertis (TBC) are shown based on capability alignment and buy vs build principles and pose the least friction from an integration and process perspective.*

**Supplier Lifecycle & Data Flow**
- SDM* (Supplier Data Management capability – application TBD)
- is responsible for validating, enriching and governing Supplier Master Data.
- SDM distributes governed supplier data to:
  - SAP ECC (current operational system)
  - SAP S/4HANA (future Business Partner model)
  - SAP Ariba for supplier collaboration, onboarding, and updates
  - iCertis for Contract Party details
  - Vistex (via SAP) for Supplier Rebate and partner function alignment

**Indirect Procurement:**
- Sourcing initiated via SAP Ariba
- Contract authoring handled in iCertis (triggered from Ariba event; may involve Legal teams)
- Approved contracts stored in iCertis, with metadata synchronised to Ariba
- POs raised via Ariba or ECC/S/4HANA depending on transition state
- Execution and financial integration performed in ECC/S/4HANA

**Direct Procurement:**
- Contract authoring via iCertis
- Supplier Master validated through SDM
- POs and goods receipts created in ECC (current) or S/4HANA (future)
- Contract compliance tracked within iCertis

Planned for deprecation | Tactical Solution | End State | Current

Current ▶
Future ▶

Internal

# S/4Hana Green Field:

# Data View

# S/4HANA GREENFIELD APPROACH – NO MIGRATION OF ECC TRANSACTIONS > SAP MDG* AS MASTER DATA HUB FOR S/4HANA AND SATELLITES

*Greenfield = A clean S/4 system with new design, new enterprise structures, governed master data and zero migration of ECC open items or historical transactions.*

**Problem / Context (short & crisp)**

- RS is implementing **S/4HANA Greenfield** with **no ECC transactional migration**.
- Only ~27% of ECC objects can be mapped to S/4; therefore, any form of mass migration is not feasible.
- To protect the new S/4 environment, master data quality must be governed *before* it enters S/4.

**Solution (MDG* Hub Concept): SAP MDG/ Informatica* will act as the central governance and distribution hub**

**Data Ingestion:**

Legacy ECC and other sources send master data to MDG (file/interface/API extract).

**Governance & Federation Rules:**

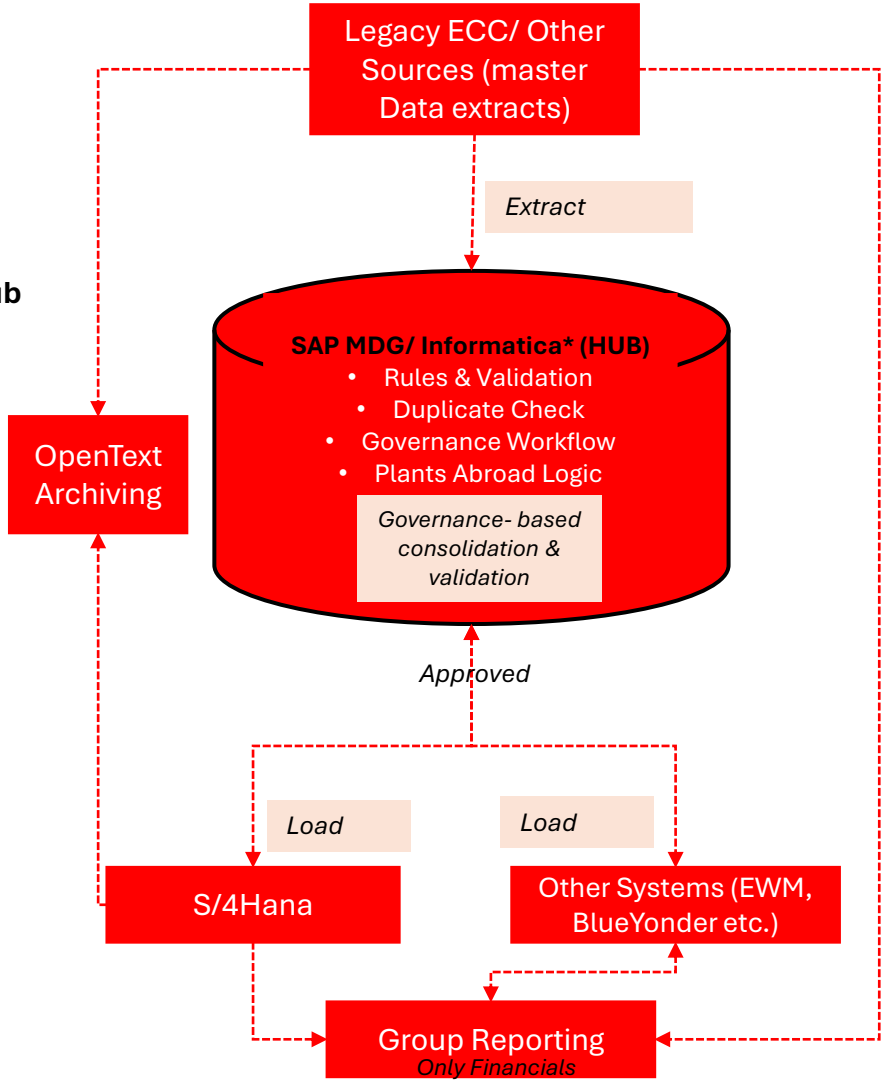
- MDG applies validation, derivation, duplication and completeness rules.
- Ensures alignment to new enterprise design (including Plants Abroad structures), chart of accounts, tax logic and global templates.
- Only data meeting governance rules is approved.

**Selective Provisioning:**

- Only fit-for-purpose, MDG-approved master data is loaded into S/4HANA.
- MDG simultaneously updates other consuming systems (Ariba, EWM, Adobe Commerce, etc.).
- No re-mapping or restructuring of ECC historical data into S/4.

**Outcome:**

- **Reduces legacy cleansing significantly by enforcing quality at MDG instead of in ECC.**
- Prevents bad data from entering S/4.
- Establishes ongoing master-data ownership and quality for future markets.
- Historic ECC data remains accessible via OpenText Archive; not transformed into S/4.
- S/4 starts clean with no legacy technical debt.




**SAP MDG* ensures only clean, governed and structurally compliant master data enters S/4HANA removing the need for legacy cleansing and avoiding migration of unsuitable ECC objects.**

* is still to be confirmed & pattern to be agreed

*S/4HANA Greenfield = Clean Start, **No ECC Migration**, Clear Separation Between OldCo (ECC) and NewCo (S/4Hana Plants Abroad)*

# Applications / Capabilities Not Yet Finalised (TBC)

# End State Logical Architecture [North Star]

# TARGET ARCHITECTURE – CORE DISTRIBUTION MASTER

Products launched for sale

Products purchased

## Capabilities

Run The Business | Manage Products | Manage Suppliers | Product Price | Market | Sell and Support | Plan and Orchestrate | Source and Return | Make and Store | Deliver and Return | Support The Business

## Value Streams

Source to Pay (Direct)

Idea to Market

Forecast to Plan

Plan to Fulfil

Source to Pay (Indirect)

Lead to Cash

Finance / R2R

Lead to Quote/Order

Order to Cash

Source to Pay (Direct)

## L1 Processes

Product & Supplier Management | Group/ Regional Finance | Inventory Management | Multiple | Indirect Procurement | Out of scope | Supply Chain Execution | Sales & Customer Service | Supply Chain Operations | Customer Experience | GSBS | Technology Mgt

| Run The Business | Manage Products | Manage Suppliers | Product Price | Market | Sell and Support | Plan and Orchestrate | Source and Return | Make and Store | Deliver and Return | Support The Business |
|---|---|---|---|---|---|---|---|---|---|---|
| Perform General Accounting & Reporting | Sourcing | Supplier Invoice Processing | Price Execution | Channel Marketing | Customer Service Management | Demand Planning | Transportation Management | Warehousing and Storage | Outbound Logistics | Invoice Customer |
| Manage Treasury Operations | Supplier Management | Revenue Accounting | Price Compliance | Customer Acquisition | | Planning Governance | Cost-to-Serve Analysis | Operational Inventory Management | Customer Orders and Returns Management | Process Adjustments |
| Manage Internal Controls | Operational Procurement | Accounts Payable | Price Assurance | Customer Insight | | Supply Planning | | Manufacturing Execution | | Process Collections |
| Manage Taxes | Supplier Invoice Processing | Supplier Invoice Processing | | Content & Experience | | Inventory Planning | | Quality Management | | Customer Credit |
| Managed Fixed Assets & Project Accounting | Accounts Payable | | | Merchanidising | | S&OP Execution | | | | |
| | Expense Re-imbursement + CC | | | | | | | | | |

## Target Architecture

- Architecture GAP
- Target Architecture
- Decision Required

| Run The Business | Manage Products | Manage Suppliers | Product Price | Market | Sell and Support | Plan and Orchestrate | Source and Return | Make and Store | Deliver and Return | Support The Business |
|---|---|---|---|---|---|---|---|---|---|---|
| Power BI | Adobe AEM | Icertis | Pricefx | GroupBy | Showpad | Mulesoft | SAP S/4 | SAP eWM | SAP S/4 | Viser |
| SAP Cloud Analytics | eClerks Product Planning Tool | Ariba | SAP S/4 | Adobe Campaign | Altify | SAP IS | SAP GBT | Exotec WCS | Metapack | PlanView |
| SAP S/4 | STIBO | SAP S/4 | | Statsig | eMatching | AWS File Transfer | | Vanderllande WCS | | M365 |
| SAP Datasphere | Expense Re-imbursement + CC | SAP Vistex | | Adobe AEM | Salesforce Servic | EDI | | Sicalis WCS | | Co-Pilot |
| Informatica MDM | SAP S/4 | DocuSign | | Tealium | Salesforce RLM | Coupa Modeller | | DAI WCS | | People Doc |
| Snowflake | SAP MDG | | | Adobe Commerce | Salesforce Sales | Informatica ETL | | KNAPP WCS | | UltiPro |
| Vertex | Semaphore | | | TreasureData | Salesforce AgentForce | SAP S/4 | | Dematic WCS | | ADP |
| Open Text | | | | Salesforce Sales | Genesys | Blue Yonder | | | | Service Now |
| SAP DRC | | | | Auth0 | SAP DPG | | | | | SAP S/4 |
| SAP GRC | | | | Content Square | Cybersource | | | | | Success Factors |
| | | | | | SAP GTS | | | | | SAP Concur |

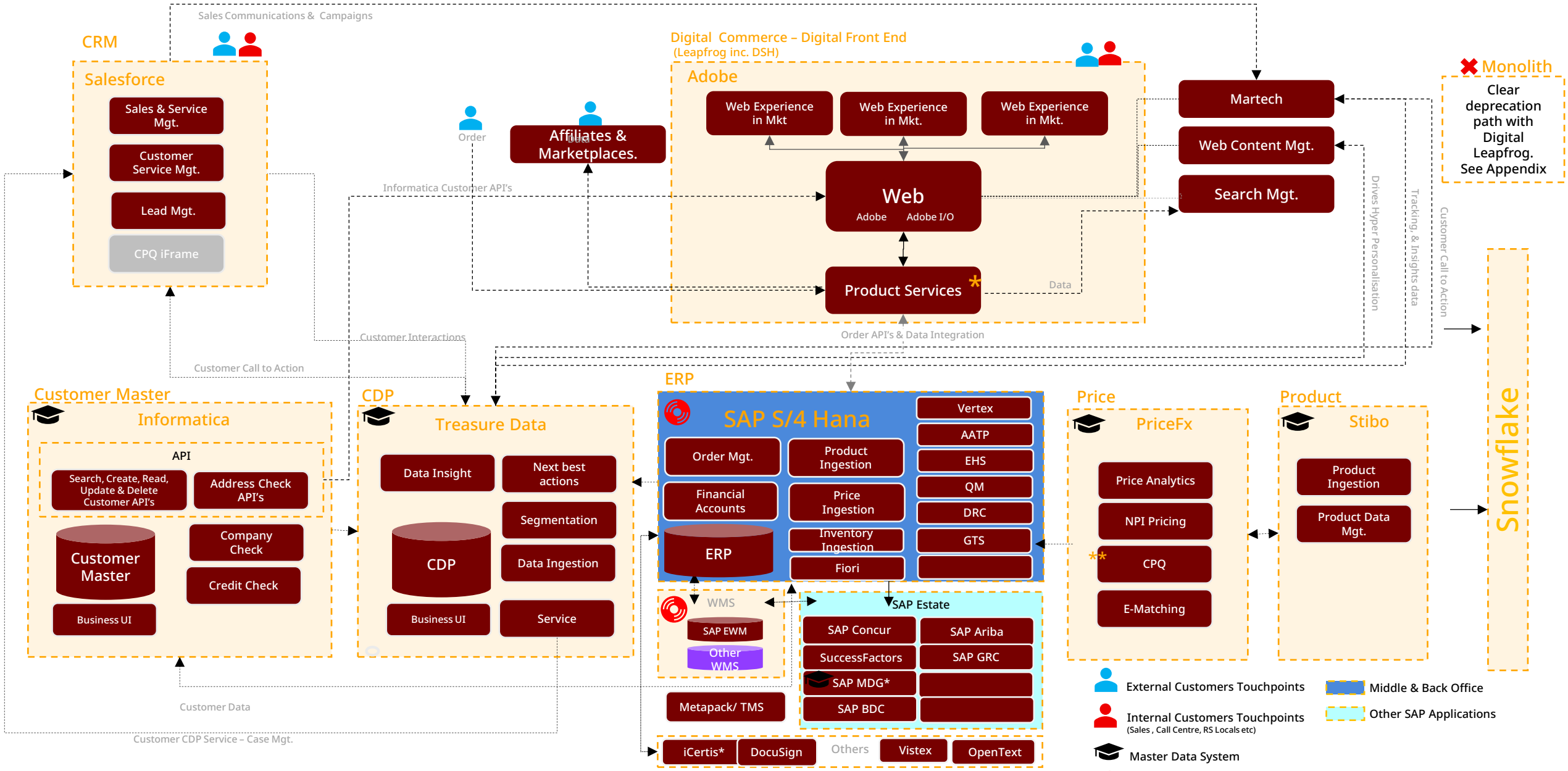
*Note: This picture is constantly evolving,*

## Eliminated Components

| Run The Business | Manage Products | Manage Suppliers | Product Price | Market | Sell and Support | Plan and Orchestrate | Source and Return | Make and Store | Deliver and Return | Support The Business |
|---|---|---|---|---|---|---|---|---|---|---|
| SAP ECC | SAP ECC | Luminance | SAP ECC | Endeca | CSOP | KONG | IBM | SAP WM | SAP ECC | SAP ECC |
| Cloudera | MAEXTRO | | | Hybris | SAP SCM | RS GRAPH | SAP ECC | Manhattan | Cloudera | Tungsten |
| SAP BW | RIVERSAND | | | JOP | TA 22 Amber Road | SAP PI | | SAP ECC WM | | |
| 6Clicks | Quadrate | | | JMOS | PriceFX/Merlin | IBM | | Manhattan | | |
| Auditboard | MDS (Americas) | | | Optimizely | QM2 | TALEND | | Demetic | | |
| SAP DI | Cloudinary | | | Cloudinary | Diabolocom | SAP ECC | | DAI | | |
| | Contentful | | | Contentful | SAP CIM | | | KNAPP | | |
| | Work Press. DMS | | | | iAdvise | | | Demetic WCS | | |
| | | | | | SalesLoft | SAP PO | | Movex | | |
| | | | | | RS SPS | | | | | |

# END – STATE LOGICAL TECHNOLOGY ARCHITECTURE

* This is a logical state and everchanging and may be correct at the time of publication (please check with the EA, for any changes




*Product Services included any integration and API's required to connect the customer front end with the enterprise backend.. More details on DSH Appendix section A1

** CPQ utilised in Salesforce and recipient of online quotes – Under review.

# Appendix

# BIG-ROCK – P2F

| # | Big Rock Capability | Why it's a Big-Rock |
|---|---|---|
| 1 | S/4HANA as Supply Chain Core | Replaces ECC, defines MRP, inventory, execution model |
| 2 | Planning → Execution Integration | Blue Yonder → S/4 MRP is a structural design |
| 3 | aATP | Changes order promising logic and customer commitments |
| 4 | Sidecar EWM | Redefines warehouse execution & stock truth |
| 5 | SAP MDG* for Product / Material | Single source of master data |
| 6 | Trade & Tax Architecture (GTS / Vertex / DRC) | Compliance model must be consistent end-to-end |
| 7 | Analytics Backbone (Snowflake + SAC) | KPI definitions & data ownership model |